Tutorial 8: PL / SQL contd.
(To be done in week8)

Tutorial Objectives:

- **Continue to learn PL / SQL**
  - **Processing with Cursors**
  - **Write Triggers**

Tutorial Tasks:

Cursors:

1 Provide a list of customers with various customer balances.
Use cursor attributes:
%FOUND or %NOTFOUND
%ROWCOUNT
%ISOPEN

a. List customers who have a balance of zero
b. List customers who have a balance of over 500

2 Provide a list of customers who have invoices.
Use cursor attributes:
%FOUND or %NOTFOUND
%ROWCOUNT
%ISOPEN

3 Provide a list of customer names as follows:
If a customer has no invoice, then the customer's name should follow with "-- no invoice"
If a customer has invoices, then the customer's name should follow with "-- has invoice"
Use cursor attributes:
%FOUND or %NOTFOUND
%ROWCOUNT
%ISOPEN

4 Rewrite at least one of the above cursors using a FOR loop (if not already used).

Triggers:

5 Write a trigger to update the customer balance in the CUSTOMER table when a new invoice record is entered. Assume that the sale is a credit sale.

Name the trigger **TRG_UPDATE_CUST_BAL_INS**.

Test the trigger using the following new INVOICE record:
8089, 1001, 17-Apr-2022, 225.40

6 Write a trigger to update the customer balance in the CUSTOMER table when an invoice is deleted.

Name the trigger **TRG_UPDATE_CUST_BAL_DEL.**

Test your trigger using the procedure **PRC_INV_DEL** (written earlier in Tutorial 7).

7 Write a trigger to record the user and date when a new customer record is added, updated, or deleted.

Note: You can store this information in another table (e.g., CUST_AUDIT). You will first need to create this table with the necessary columns.